IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

CONTROL SYSTEMS

Wednesday, 30 April 2025, 2:30 PM

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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CONTROL SYSTEMS – 2024/2025

Information for candidates:

Remark for the Students: the information provided here is much more than needed to answer the Exam Paper Questions; that is, the presence of a specific formula does not necessarily mean that this formula is required to answer the questions.

- **Bode Plots to answer Question 3**

Remark for the Students: In case drawing Bode plots is deemed helpful or necessary to answer Question 3), an additional answer book is provided that contains a few clean semi-logarithmic charts that should be used and should be attached to the main Answer Book.

Information for candidates (continued):

Main Laplace Transforms and General Properties

$f(t)$	$F(s)$
$1(t)$	$\frac{1}{s}$
$t \cdot 1(t)$	$\frac{1}{s^2}$
$t^2 \cdot 1(t)$	$\frac{2}{s^3}$
$e^{\alpha t} \cdot 1(t)$	$\frac{1}{s-\alpha}$
$t \cdot e^{\alpha t} \cdot 1(t)$	$\frac{1}{(s-\alpha)^2}$
$\sin(\omega t) \cdot 1(t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos(\omega t) \cdot 1(t)$	$\frac{s}{s^2+\omega^2}$
$t \cdot \sin(\omega t) \cdot 1(t)$	$\frac{2\omega s}{(s^2+\omega^2)^2}$
$t \cdot \cos(\omega t) \cdot 1(t)$	$\frac{s^2-\omega^2}{(s^2+\omega^2)^2}$
$e^{\sigma t} \cdot \sin(\omega t) \cdot 1(t)$	$\frac{\omega}{(s-\sigma)^2+\omega^2}$
$e^{\sigma t} \cdot \cos(\omega t) \cdot 1(t)$	$\frac{s-\sigma}{(s-\sigma)^2+\omega^2}$
$t \cdot e^{\sigma t} \cdot \sin(\omega t) \cdot 1(t)$	$\frac{2\omega(s-\sigma)}{\left[(s-\sigma)^2+\omega^2\right]^2}$
$t \cdot e^{\sigma t} \cdot \cos(\omega t) \cdot 1(t)$	$\frac{(s-\sigma)^2-\omega^2}{\left[(s-\sigma)^2+\omega^2\right]^2}$

Table 1 Time-functions and corresponding Laplace transforms

$\mathscr{L}[f(t)+g(t)]$	$F(s)+G(s)$
$\mathscr{L}\left[\int_0^t f(\tau)d\tau\right]$	$\frac{1}{s}F(s)$
$\mathscr{L}\left[\dot{f}(t)\right] = \mathscr{L}\left[\frac{d}{dt}f(t)\right]$	$sF(s)-f(0)$
$\mathscr{L}\left[\ddot{f}(t)\right] = \mathscr{L}\left[\frac{d^2}{dt^2}f(t)\right]$	$s^2F(s)-sf(0)-\dot{f}(0)$
$\mathscr{L}\left[\frac{d^3}{dt^3}f(t)\right]$	$s^3F(s)-s^2f(0) -s\dot{f}(0)-\ddot{f}(0)$
$\mathscr{L}\left[e^{kt}f(t)\right]$	$F(s-k)$
$\mathscr{L}[f(t-t_0)]$	$e^{-st_0}F(s)$
$\mathscr{L}[f(at)], a \in \Re, a>0$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
$\mathscr{L}[t^n f(t)]$	$(-1)^n\frac{d^n}{ds^n}F(s)$
$\mathscr{L}[f(t)*g(t)] =$	
$= \mathscr{L}\left[\int_0^t f(\tau)g(t-\tau)d\tau\right]$	$F(s)G(s)$
If $\deg[N(s)] < \deg[D(s)]$ with $F(s)=N(s)/D(s)$	$f(0)=\lim_{s\to\infty}F(s)$
If all poles of $F(s)$ have negative real part except (possibly) one pole in $s=0$	$\lim_{t\to\infty}f(t)=\lim_{s\to 0}sF(s)$

Table 2 Main properties of Laplace tranforms

1. Consider the block scheme shown in Figure 1.1:

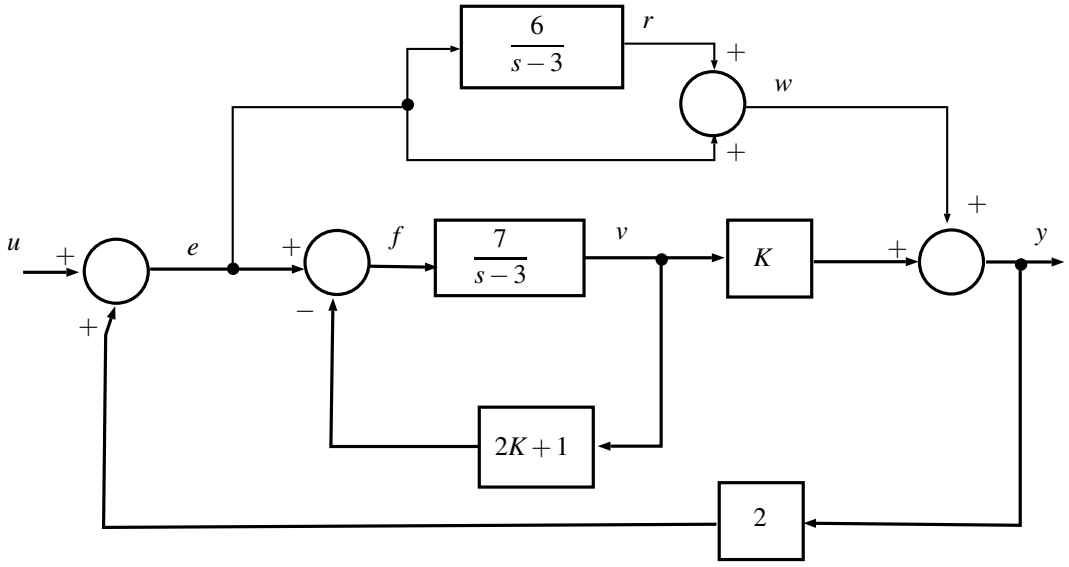


Figure 1.1 Block scheme for Question 1.

where, with some abuse of notation, we represent the inputs and outputs of each block in the time-domain instead of in the Laplace domain. Moreover, $K \in \mathbb{R}$ is a generic scalar, and the transfer functions in all blocks have no hidden common factors in their numerators and denominators.

a) Determine a state-space description of the dynamic system shown in Figure 1.1 of the form:

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases}$$

where $x \in \mathbb{R}^n$ is a state vector to be suitably defined, $\dot{x}$ denotes its time-derivative, u denotes the input vector, y denotes the output vector, and A, B, C, D are suitable matrices to be determined.

[7 Marks]

b) Establish the stability properties of the dynamic system shown in Figure 1.1 as a function of the scalar $K \in \mathbb{R}$.

[3 Marks]

c) By just inspecting the block scheme shown in Figure 1.1, find (if possible) a value of the scalar $K \in \mathbb{R}$ such that the whole dynamic system is unstable. Discuss your findings with respect to your answer to Question 1-b).

[10 Marks]

d) Setting $K = 0$ determine the transfer function $G_{u,y}(s)$ from the input $u(t)$ to the output $y(t)$.

[10 Marks]

e) Setting $K = 1$ determine the transfer function $G_{u,y}(s)$ from the input $u(t)$ to the output $y(t)$.

[10 Marks]

2. a) Consider the nonlinear system described by the state equations:

$$\begin{cases} \dot{x}_1(t) &= 4\left[x_1(t)\right]^4 + 2\left[x_1(t)\right]^2 + 2x_2(t) - u(t) \\ \dot{x}_2(t) &= -2\left[x_1(t)\right]^4 - x_2(t) - 2.5\,u(t) \end{cases}$$

i) Determine all possible equilibrium states corresponding to the constant input $u(t) = \bar{u} = 3, \forall t$.

[3 Marks]

ii) Determine the state equations of the linearised system on the equilibrium state $\bar{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ corresponding to the constant input $u(t) = 0, \forall t$.

[3 Marks]

iii) Analyse the stability of $\bar{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ as an equilibrium state of the above nonlinear system.

[3 Marks]

b) Consider the linear system described by the state equations:

$$\begin{cases} \dot{x}_1(t) &= -x_1(t) + \dfrac{1}{2}\,x_2(t) + u(t) \\ \dot{x}_2(t) &= -2x_1(t) + x_2(t) \\ y(t) &= -x_2(t) \end{cases}$$

Set the initial state to $x(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and the input as

$$u(t) = \mathbf{1}(t)$$

where $\mathbf{1}(t)$ denotes the unit step function.

Determine the analytical expression of the output function $y(t), t \in \mathbb{R}$.

[11 Marks]

3. Consider the closed-loop control system represented by the block scheme shown in Figure 3.1:

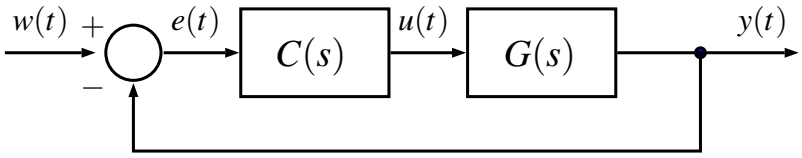


Figure 3.1 Block scheme for Question 3.

where, with some abuse of notation, we represent the inputs and outputs of each block in the time-domain rather than the Laplace domain. Moreover

$$G(s) = 4.8 \cdot \frac{1+0.15s}{1-0.15s} e^{-Ts}$$

and T denotes a delay of T sec.

a) Setting $T = 0$ seconds, design a controller $C(s)$ such that:

i) The crossover frequency satisfies $\omega_c \geq 10$ rad/seconds.

ii) The phase margin φ_m satisfies $\varphi_m \geq 60^\circ$.

iii) The controller $C(s)$ is a strictly proper dynamic system.

If needed or deemed useful, use the semi-logarithmic charts provided in the Additional Answer Book.

[20 Marks]

b) Again, setting $T = 0$ seconds, and using the controller designed in your answer to Question 3-a), set the reference input $w(t) = (1 - e^{-t}) \cdot \mathbf{1}(t)$, where $\mathbf{1}(t)$ denotes the unit step function.

i) Determine the analytical expression of the output function $y(t), t \in \mathbb{R}$ under the action of the closed-loop controller.

[6 Marks]

ii) Sketch the diagrams of the reference input function $w(t) = (1 - e^{-t}) \cdot \mathbf{1}(t)$ and of the output function $y(t), t \in \mathbb{R}$ determined in the answer to Question 3-b)-i).

[3 Marks]

iii) Compare the diagrams drawn in the answer to Question 3-b)-ii) and comment on your findings.

[3 Marks]

c) Now, suppose that $0 < T \leq \overline{T}$, where $\overline{T} > 0$ is an upper bound on the delay T.

Consider the controller $C(s)$ designed in your answer to the Question 3-a) for $T = 0$ seconds and denote by φ_m the corresponding phase margin.

Determine a suitable upper bound $\overline{T} > 0$ such that the decrease $\Delta\varphi_m$ of the phase margin φ_m caused by any time-delay T with $0 < T_d \leq \overline{T}$ satisfies:

$$|\Delta\varphi_m| \leq 20^\circ$$

[8 Marks]